

Set covering

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Practice quiz

You need to recruit software developers from the following team of outstanding computer scientists: Alan, Bjarne, Tim, and Donald. Each of them is expert in at least one programming language:

- Alan in C and C++,
- Bjarne in C++ and Java,
- Tim in C++, Ruby and Python,
- Donald in C and Java.

Your job is to hire a team of experts, with the following requirements:

1. each language must be covered, in the sense that, for each language, there should be at least one person in the team mastering it,
2. the team must be as small as possible.

Model that set covering problem as an integer optimization problem.